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Multimodal Clustering for Disease-Phenotype Discovery with Chest X-Ray and Tabular Features

– Data Analysis and Knowledge Discovery Report –

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Abstract

Text of abstract. Short info about:

- project relevance/importance,
- intelligent methods used for solving,
- data involved in the numerical experiments;
- conclude by the the results obtained.
- Please add a graphical abstract of your work.

Remind that a good report should:

- be fun to read with many figures and visualizations;
- be easy to follow even for AI/ML novice;
- clearly convey the potential of AI/ML to the application domain;
- around 10 minutes to read (although this is not a hard constraint).

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Chapter 1

Introduction

1.1 What? Why? How?

1.1.1 Problem Description

Modern medical datasets increasingly contain heterogeneous data modalities, such as medical images (e.g., chest X-rays) alongside structured clinical variables (e.g., demographics, observations, and diagnostic labels). Traditional analysis pipelines typically treat these modalities independently, limiting their ability to capture complex relationships between visual and clinical features. As a result, discovering meaningful patient subgroups or disease phenotypes remains a challenging task.

In this work, we address the problem of *multimodal clustering for disease phenotype discovery*, where the goal is to group patients into clinically meaningful clusters based on both imaging and tabular data. Specifically, we aim to identify latent disease patterns in the CheXpert dataset by jointly leveraging chest X-ray images and associated metadata.

1.1.2 Motivation and Importance

A *disease phenotype* is defined as a set of observable characteristics of a disease, including imaging findings, clinical features, and disease progression patterns, that collectively describe distinct manifestations of the underlying pathology.

In a real-world clinical setting, radiologists analyze chest X-rays together with patient information such as age, sex, and medical history to diagnose conditions like pneumonia, cardiomegaly, or pleural effusion. However, patients with the same diagnosis label can exhibit very different visual patterns and clinical characteristics. For example, two patients diagnosed with pneumonia may have different underlying causes, severity levels, or responses to treatment, which are not explicitly captured in

standard labels.

Identifying disease phenotypes is important *for improving diagnosis, prognosis, and personalized treatment*. Many diseases, particularly in radiology, exhibit high heterogeneity, meaning that patients with similar diagnoses may present different visual patterns or clinical profiles. Unsupervised clustering can reveal hidden subtypes that are not captured by predefined labels.

In real-world clinical settings, physicians rely on both imaging data and patient information to make decisions. Therefore, integrating these modalities computationally can lead to more accurate and clinically relevant insights. Multimodal clustering has the potential to:

- Discover novel disease subtypes beyond existing annotations
- Improve patient stratification for treatment planning
- Support clinical decision-making through data-driven insights

1.1.3 Proposed Approach

To address this problem, we propose a *multimodal clustering framework* that combines features extracted from medical images with structured tabular data. Image features are obtained using deep learning models (e.g., convolutional neural networks or pretrained encoders), while tabular features are preprocessed and encoded appropriately. These representations are then fused into a shared latent space, where clustering algorithms (e.g., K-means, spectral clustering, or deep clustering methods) are applied to identify disease phenotypes.

1.1.4 Relation to Prior Work

Recent advances in multimodal learning have demonstrated the effectiveness of combining imaging and clinical data for medical tasks. Several studies have explored joint representation learning using deep neural networks, contrastive learning, or autoencoder-based approaches. However, most prior work focuses on supervised prediction tasks, while unsupervised multimodal clustering for phenotype discovery remains relatively underexplored.

1.1.5 Contributions and Expected Results

In this project, we aim to:

- Develop a multimodal representation learning pipeline for chest X-ray data
- Apply clustering techniques to identify disease phenotypes

- Evaluate the quality and clinical relevance of the discovered clusters

We expect that integrating both image and tabular data will lead to more coherent and clinically meaningful clusters compared to unimodal approaches. The results will provide insights into disease heterogeneity and demonstrate the potential of multimodal learning in medical data analysis.

1.2 Paper structure and original contribution(s)

The rest of this paper is organized as follows. Chapter 2 presents the related work, focusing on multimodal learning, medical image representation learning, multimodal fusion, and phenotype discovery methods relevant to chest X-ray analysis. Chapter 3 describes the investigated approach, including the datasets, exploratory data analysis, preprocessing steps, image feature extraction, multimodal feature fusion, dimensionality reduction, clustering algorithms, and validation methodology. The experimental results are then analyzed using internal clustering metrics, t-SNE visualizations, and disease-label enrichment through prevalence and lift scores. Finally, Chapter 5 summarizes the main findings, discusses the strengths and limitations of the proposed approach, and outlines possible directions for future work.

The original contribution of this work consists of designing and evaluating a multimodal clustering pipeline for disease-phenotype discovery from chest X-ray data. Unlike standard supervised approaches that aim to predict predefined disease labels, this project focuses on discovering hidden patient subgroups by combining visual features extracted from radiographs with structured tabular information. The study compares several clustering algorithms, including K-Means, HDBSCAN, Hierarchical Clustering, Spectral Clustering, and Gaussian Mixture Models, and evaluates their behavior across multiple experimental settings. An additional contribution is the clinical interpretation of the discovered clusters using disease prevalence and lift scores, which provides insight into how the clusters relate to possible disease phenotypes. The use of both CheXpert and NIH ChestX-ray14 also allows the robustness of the proposed pipeline to be examined across different chest X-ray datasets.

1.3 Problem definition

The problem addressed in this project is the discovery of disease phenotypes from chest X-ray data using multimodal clustering. Medical datasets often contain different types of information, such as radiographic images, patient metadata, and diagnostic labels. In this work, these data sources are combined in order to group similar patients or images into meaningful clusters.

The goal is to identify hidden patterns in the data without using predefined target classes. Instead of predicting a known disease label, the clustering algorithms try to discover groups of samples that share similar visual and clinical characteristics. These groups can be interpreted as possible disease phenotypes.

Formally, each sample can be represented as:

$$x_i = [v_i, t_i],$$

where v_i represents the features extracted from the chest X-ray image and t_i represents the tabular clinical features. The output of the clustering process is a cluster label:

$$c_i \in \{1, 2, \dots, K\},$$

where c_i indicates the group assigned to sample i . For density-based methods such as HDBSCAN, some samples may also be labelled as noise if they do not clearly belong to any cluster.

This problem requires intelligent algorithms because chest X-ray data are complex, high-dimensional, and noisy. Images contain visual patterns that are difficult to describe manually, while medical labels may be imbalanced, uncertain, or incomplete. Simple rule-based methods are not suitable for this type of problem because they cannot capture the hidden relationships between image features and clinical information.

Several clustering methods can be applied to this task. K-Means is simple and efficient, but it assumes compact clusters. Hierarchical Clustering can show relationships between groups, but it may be sensitive to the chosen linkage method. HDBSCAN can detect dense groups and outliers, but it may leave many samples unassigned. Spectral Clustering can capture non-linear structures, while Gaussian Mixture Models provide probabilistic cluster assignments.

The importance of this problem comes from the fact that patients with the same diagnosis may still present different visual patterns or clinical characteristics. By discovering subgroups automatically, multimodal clustering can help reveal disease heterogeneity and support a better understanding of patient profiles in chest X-ray datasets.

Chapter 2

Related work

Recent work on combining chest X-ray images with structured clinical information falls into three main categories: **multimodal representation learning**, **multimodal fusion for predictive modeling**, and **explicit unsupervised phenotype discovery**. In the first category, methods such as MedAug [8], Multimodal MSN [7], and Symile [3] focus on learning joint embeddings from multiple modalities. In the second category, approaches such as MedFuse [4] and HyperFusion [2] develop fusion architectures for clinical prediction, but their learned feature spaces are also relevant for clustering. In the third category, M-ClustEHR [1] is particularly important because it directly addresses multimodal clustering and phenotype discovery. We also discuss Diff3M [6], as it shows how multimodal representations can help identify atypical or rare patterns in chest X-rays. These papers were selected because they are the most relevant to our problem setting and together provide a representative overview of the main methodological directions related to multimodal disease phenotype discovery.

The authors of [8] introduced MedAug as a contrastive learning framework that incorporates patient metadata into the construction of positive pairs, instead of relying only on two augmentations of the same image.

Methodologically, the main idea is to enrich contrastive self-supervision by using metadata such as patient identity, study identity, and laterality constraints when defining positive pairs. The framework is implemented on top of MoCo v2, and the authors also explored metadata-informed hard negative selection, although this did not lead to clear improvements. The learning objective remains based on the standard InfoNCE-style contrastive loss, but the augmentation strategy is extended through a metadata-dependent enhanced augmentation set.

For experiments, the authors used CheXpert, which contains 224,316 chest X-ray images from 65,240 patients with labels for 14 observations. Pretraining was performed on the unlabeled data,

while fine-tuning used only 1% labeled data for pleural effusion classification. The evaluation was carried out on a test set of 500 labeled images from 500 studies excluded from training. The reported MoCo v2 setup uses a ResNet-18 backbone, batch size 16, temperature 0.2, and 20 training epochs.

In terms of results, the best positive-pair strategy, namely pairing images from the same patient and study across lateralities, produced a substantial improvement over the ImageNet-pretrained baseline. In the abstract, the authors report a +14.4% mean AUC gain. They also compared different negative-pair strategies, but the standard default negatives performed as well as, or better than, the metadata-driven alternatives.

A limitation of the work is that it evaluates performance through downstream pleural effusion classification, not through any direct measure of cluster quality or phenotype discovery. In addition, the lack of gains from metadata-driven hard negatives suggests that the main benefit comes from positive-pair design. This leaves open the broader question of which kinds of clinical metadata truly correspond to a shared phenotype and which may introduce spurious similarity.

Hayat, Geras, and Shamout [4] proposed MedFuse as a multimodal framework for combining chest X-ray images with clinical time-series information.

The method follows a two-stage design. First, modality-specific encoders are pretrained separately for the EHR time-series data and the chest X-ray images. Then, the learned latent representations are fused using an LSTM-based fusion module that treats modality embeddings as a sequence of tokens. Concretely, the time-series encoder is a stacked LSTM that produces a 256-dimensional representation, while the image encoder is a ResNet-34 that outputs a 512-dimensional embedding, later projected down to 256 dimensions before fusion. The final fusion module is a single-layer LSTM with hidden size 512, and the model is trained using binary cross-entropy, with evaluation in terms of AUROC and AUPRC.

The experiments link MIMIC-IV clinical time series with MIMIC-CXR chest X-ray images. The clinical data follow a standard ICU benchmark setting, using 17 variables sampled every 2 hours, then discretized and standardized, which yields a 76-dimensional feature vector per time step after one-hot encoding. Splits are defined at the patient level using a 70% / 10% / 20% train/validation/test partition. The study also considers a paired and partially paired multimodal setting; for example, in the phenotyping task, the training set contains 8,056 paired samples with chest X-rays out of 42,628 total samples. For the image modality, the reported MIMIC-CXR split sizes are 325,188 / 15,282 / 36,625 images.

Experimentally, MedFuse achieved the strongest performance among the compared fusion baselines

on the paired test set. For phenotype classification, the best configuration reported 0.770 AUROC and 0.481 AUPRC, while for in-hospital mortality prediction it achieved 0.865 AUROC and 0.594 AUPRC. Confidence intervals were estimated bootstrap, and the authors also identified an "optimal" amount of unimodal data during training, namely 10% for mortality prediction and 20% for phenotyping.

Its main limitation is that the work focuses on prediction tasks, specifically mortality and phenotype classification, rather than unsupervised phenotype discovery. Nevertheless, it offers a principled way to produce multimodal latent representations, and one open question is whether the most cluster-informative features lie in the pre-fusion modality embeddings or in the final fused hidden state.

Shurrab, Guerra-Manzanares, and Shamout [7] proposed an extension of masked siamese self-supervised learning for chest X-rays that integrates structured EHR information.

The paper extends Masked Siamese Networks (MSN) by introducing static EHR features during pretraining. The architecture contains two visual branches, one for the anchor view and one for the target view, together with an EHR encoder and a fusion projection module. The EHR vector is encoded, concatenated with the chest X-ray latent representation, and then projected into the same space as the target representation. The study explores three groups of structured variables, namely demographics, scan metadata, and inpatient stay information.

The pretraining and internal validation experiments are carried out on MIMIC-CXR linked with structured information from MIMIC-IV. The authors report the size of MIMIC-CXR as 377,110 radiographs from 227,835 studies, and they follow patient-level splits consistent with prior work, with 325,188 images for training, 15,282 for validation, and 36,625 for testing. External validation is then performed on CheXpert and NIH-14, with CheXpert used as an external test set and NIH-14 using its own train/validation/test protocol because of label differences. In terms of preprocessing, the augmentations include resized cropping and horizontal flipping, while color jitter and Gaussian blur are omitted because they are not appropriate for this domain.

The paper reports that multimodal MSN improves linear evaluation AUROC over vanilla MSN and several other self-supervised baselines. According to the abstract, the gain is around +2% AUROC over standard MSN and +5–8% over other baselines. The full paper provides AUROC and AUPRC values with confidence intervals for external evaluation on CheXpert and NIH-14.

A notable limitation is that the gains are most visible in linear evaluation, whereas end-to-end fine-tuning shows weaker or inconsistent improvement. The authors suggest that removing EHR information during downstream fine-tuning may dilute the advantage learned during pretraining. They also observe that combining multiple EHR feature groups can sometimes hurt performance, which

raises broader questions about feature interaction and the best level at which structured data should be fused.

Saporta, Puli, Goldstein, and Ranganath [3] introduced Symile at NeurIPS 2024 as a general framework for representation learning across more than two modalities. Both code and datasets were released by the authors.

The key methodological contribution is a contrastive objective based on total correlation, which generalizes contrastive learning beyond pairwise alignment and aims to capture higher-order and conditional dependencies among multiple modalities. Importantly, the method remains architecture-agnostic: each modality can be encoded independently and mapped into a shared space, while training is driven by the Symile loss averaged across different anchor-modality choices. The paper also discusses practical approximations for negative sampling and the challenges of scaling contrastive learning to settings with more than two modalities.

For the clinical case study, the authors introduce Symile-MIMIC, a dataset combining chest X-rays, ECGs, and blood laboratory values from MIMIC-IV, MIMIC-IV-ECG, and MIMIC-CXR-JPG. The dataset contains 11,622 admissions, partitioned into a development set comprising 95% of patients and a test set containing the remaining 5%, with no patient overlap. The documentation also specifies preprocessing steps, including image scaling so that the smaller edge equals 320 pixels, square cropping, and ImageNet normalization for the chest X-rays. Only AP/PA views are retained. Laboratory data are encoded as 50 percentile-standardized values plus 50 missingness indicators, which is a particularly relevant design choice for multimodal clinical learning.

The evaluation task is formulated as zero-shot retrieval. Given an ECG-plus-labs query, the model must retrieve the corresponding chest X-ray from a set of ten candidates, one of which is correct. On this task, the paper reports a mean retrieval accuracy of 0.435 ± 0.007 standard error, compared with 0.387 ± 0.003 for a CLIP-style baseline, which corresponds to an approximately 12.5% relative improvement.

A limitation is that Symile is best motivated when there are meaningful higher-order dependencies across several modalities. If the non-image branch contains only weak metadata, it is not clear whether the extra complexity of the method is justified. The paper also points toward future work on more memory-efficient objectives and improved negative sampling schemes for many-modality settings.

Duenias, Nichyporuk, Arbel, and Riklin Raviv [2] proposed HyperFusion as a multimodal fusion method that conditions image processing on tabular variables through a hypernetwork.

Instead of aligning modalities in a shared embedding space, HyperFusion uses a hypernetwork to

generate or modulate the weights and biases of selected layers in the primary image network. In this way, image feature extraction becomes conditioned on tabular attributes at inference time. The authors argue that this is more appropriate when the structured variables are not semantically paired with image content in the way required by CLIP-style methods, for instance when the structured variables are attributes such as sex or temperature rather than another direct sensory modality.

The experiments are performed on brain MRI tasks, specifically brain age prediction conditioned on sex and multiclass Alzheimer's disease classification among CN, MCI, and AD categories using data derived from ADNI. The paper reports population statistics such as cohort sizes, ages, and sex distributions, and it preprocesses the MRI volumes by cropping 3D subimages of size $64 \times 96 \times 64$ around the hippocampal regions.

Results are reported using several metrics. For the Alzheimer's disease classification task, the paper includes macro AUC, macro F1, balanced accuracy, PRC, and class-specific true positive rates, summarized in Table 3 as means and standard deviations over multiple splits and initializations. For brain age prediction, the paper reports comparisons in terms of MAE, including the relevant figures and ablations.

The main limitation is that the method has not been validated on CXR data, so its transferability to this domain remains uncertain. Another open issue concerns which layers should be hyper-conditioned, since the paper itself includes ablations around hyperlayer selection.

Kim, Wang, Ahn, Choi, Zhou, and Hong [6] proposed Diff3M as a multimodal anomaly detection framework for chest X-rays that integrates structured EHR information.

The method is trained only on normal chest X-rays and performs anomaly detection by reconstructing a normal-like image and measuring deviations. Structured EHR variables are incorporated through an Image-EHR Cross Attention (IECA) module, while Pixel-level Checkerboard Masking (PCM) is used to provide a fixed masking scheme during training. The objective combines a diffusion-style noise prediction loss with a weighted reconstruction loss, both defined using mean squared error.

The experiments are carried out on CheXpert and MIMIC-CXR/MIMIC-IV in an unsupervised anomaly detection setting, where "No Finding" cases define the normal class and all others are treated as anomalies. On the CheXpert side, the structured inputs include sex, age, and AP/PA projection view, while for MIMIC the model additionally uses BMI, blood pressure, height, and weight, provided that those measurements are available within three months of image acquisition. The paper reports 16,969 normal CheXpert samples for training and a test set containing 26 normal and 176 abnormal cases. For MIMIC, it reports 29,310 normal training samples, although the rest of the details continue

beyond the extracted portion.

In terms of results, Diff3M is compared against feature-based baselines such as PatchCore, RD4AD, and MambaAD, as well as the diffusion-based baseline DiAD. On CheXpert, the variant labeled Ours_{demo} achieves AUROC 0.664 and AUPRC 0.931, improving over DiAD's AUPRC 0.917. On MIMIC, adding richer EHR variables in Ours_{demo,ehr} yields AUROC 0.617 and AUPRC 0.821.

The main limitation is that Diff3M is not itself a clustering method. Although it learns representations that may be useful for grouping anomalies, there is no guarantee that those latent features are optimally structured for clustering. A second drawback is the computational cost of diffusion-based training, which makes the method more suitable as a specialized tool for rare phenotype mining than as the main representation backbone.

Bampa, Miliou, Jovanovic, and Papapetrou [1] introduced M-ClustEHR as a framework explicitly aimed at unsupervised phenotype discovery. This makes it one of the most directly relevant papers conceptually, even though it does not include imaging.

The method learns a shared latent representation through a Multimodal Autoencoder (MMAE). Different modalities are first encoded separately, after which the latent representations are combined into a unified feature space on which clustering is performed. The paper includes separate autoencoders for continuous variables and longitudinal signals, with the longitudinal branch implemented as a 3-layer GRU autoencoder. The resulting latent vectors are concatenated into a joint representation that serves as the input to the clustering stage.

The experiments use raw MIMIC-III data, with comorbidities represented according to the Elixhauser index. Two sepsis cohorts are considered, namely the Sepsis-3 cohort and the Angus cohort. In the extracted table, the authors report cohort sizes such as 5,784 patients for Sepsis-3 and 2,867 patients for the Angus cohort, together with statistics related to length of stay and mortality.

The evaluation is centered on clustering quality and stability, rather than downstream prediction. The authors report internal clustering metrics such as Silhouette score and Davies Bouldin index for several clustering algorithms, including K-Prototypes, ensemble clustering, HDBSCAN, and hierarchical clustering. They also define an explicit cluster stability score based on bootstrap sampling and alignment of cluster labels, which provides an important methodological contribution for phenotype discovery studies.

The main limitation is that the study is still EHR-centric, so the question of how the discovered phenotypes relate to imaging patterns remains open. In addition, the PDF version accessed in the earlier extraction was labeled as a preprint copy, so exact claims and numbers should ideally be verified

against the final journal version before inclusion in a formal thesis.

Paper (year)	Primary goal	Method type	Fusion strategy	Clustering component	Datasets (modalities)	Top metrics
Vu et al. (2021) [8]	Metadata-aware CXR representation learning	Contrastive SSL (MoCo v2)	Metadata-defined positives	No; embeddings only	CheXpert (CXR + metadata)	+14.4% mean AUC vs ImageNet
Hayat et al. (2022) [4]	Multimodal fusion for clinical prediction	Supervised fusion	LSTM over modality tokens	No; fused latent can be clustered	MIMIC-IV + MIMIC-CXR (EHR + CXR)	0.770 AUROC / 0.481 AUPRC; 0.865 / 0.594
Shurrab et al. (2024) [7]	Multimodal SSL for CXR representation learning	Transformer SSL (MSN)	Latent concat + projection	Implicit; prototype space in SSL	MIMIC-CXR + MIMIC-IV; CheXpert/NIH-14	AUROC/AUPRC gains with CIs
Saporta et al. (2024) [3]	Shared representation learning for 3+ modalities	Contrastive SSL (total correlation)	Joint multimodal contrast	No; shared embeddings only	Symile-MIMIC (CXR + ECG + labs)	Retrieval acc.: 0.435 vs 0.387
Duenias et al. (2025) [2]	Tabular-conditioned image representation learning	Hypernetwork conditioning	Tabular $\rightarrow$ image weights	No; conditioned features only	ADNI (MRI + tabular)	Macro AUC/F1/BA/PRC; MAE
Kim et al. (2025) [6]	Multimodal anomaly detection in CXR	Diffusion UAD + EHR	Cross-attention conditioning	No; anomaly latents can be clustered	CheXpert; MIMIC-CXR/IV	CheXpert AUPRC up to 0.931
Bampa et al. (2024) [1]	Multimodal phenotype discovery	Multimodal AE + clustering	Shared latent (MMAE)	Yes; clustering on shared latent space	MIMIC-III sepsis cohorts	Silhouette/DB + stability

Table 2.1: Summary of multimodal learning approaches in medical imaging and their relevance to clustering and representation learning.

Chapter 3

Investigated approach

3.1 Proposed approach - methodology

The proposed approach is based on a multimodal clustering pipeline designed to discover possible disease phenotypes from chest X-ray data. The input data consist of two main types of information: visual data, represented by chest X-ray images, and structured tabular data, represented by patient-related and diagnostic features such as age, sex, and radiological observations. First, visual features are extracted from the X-ray images using a pretrained deep learning model, while the tabular features are cleaned, encoded, and normalized. These two feature types are then fused into a single multimodal representation by concatenating the image embeddings with the processed tabular variables. The resulting feature space is used as input for several clustering algorithms in order to group similar samples together. Each discovered cluster is interpreted as a potential disease phenotype, meaning a subgroup of patients or images that share similar visual and clinical characteristics.

3.1.1 Data

The dataset used in this study is the CheXpert dataset [5], a large-scale public dataset of chest radiographs developed by Stanford University. It contains 224,316 chest X-ray images from 65,240 patients, along with associated structured metadata such as patient demographics (e.g., age and sex) and radiology observations.

Each image is labeled for the presence of 14 different observations (including conditions such as Atelectasis, Cardiomegaly, Edema, and Pleural Effusion), extracted automatically from radiology reports using natural language processing. The labels can take values of positive (1), negative (0), or uncertain (-1), reflecting the ambiguity often present in clinical interpretations.

The dataset includes both frontal and lateral chest X-rays, though many studies focus on frontal

images due to their higher availability and consistency. CheXpert is particularly well-suited for multi-modal analysis, as it combines high-dimensional image data with structured clinical variables, making it an appropriate choice for tasks such as disease phenotype discovery through clustering.

For the final experiment, an additional dataset was used: the NIH ChestX-ray14 dataset¹. This is a large-scale public chest X-ray dataset released by the National Institutes of Health, containing frontal-view chest radiographs annotated with 14 thoracic disease labels. These labels include conditions such as Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, and others. Similar to CheXpert, the dataset provides multi-label disease annotations, meaning that each image may be associated with one or more pathological findings.

The NIH ChestX-ray14 dataset was included in the last experiment in order to evaluate the clustering pipeline on an additional radiology dataset and to test whether the discovered disease phenotype patterns remain meaningful beyond the CheXpert data. Since it also contains chest X-ray images and structured diagnostic labels, it is suitable for validating the proposed multimodal clustering approach in a broader experimental setting.

3.1.2 Exploratory Data Analysis (EDA)

An exploratory analysis was conducted to better understand the structure, quality, and characteristics of the CheXpert dataset prior to preprocessing and modeling. The dataset consists of chest X-ray studies annotated with 14 radiological observations, along with patient metadata such as *age* and *sex*.

The distribution of disease labels from Figure 3.1 highlights a significant class imbalance across observations. Conditions such as *Support Devices*, *Lung Opacity*, and *Pleural Effusion* are among the most frequently occurring positive findings, whereas others like *Fracture*, *Pneumothorax*, and *Lung Lesion* are relatively rare.

A substantial number of missing values is observed across multiple features according to Figure 3.2, particularly for labels such as *Pleural Other*, *Lung Lesion*, and *Fracture*. This sparsity reflects the clinical reality where not all conditions are explicitly mentioned in radiology reports.

The disease co-occurrence matrix from Figure 3.3 further reveals meaningful relationships between conditions. Strong co-occurrence patterns are observed between:

- *Edema* and *Pleural Effusion*
- *Consolidation* and *Pneumonia*
- *Atelectasis* and *Consolidation*

¹<https://www.kaggle.com/datasets/nih-chest-xrays/data>

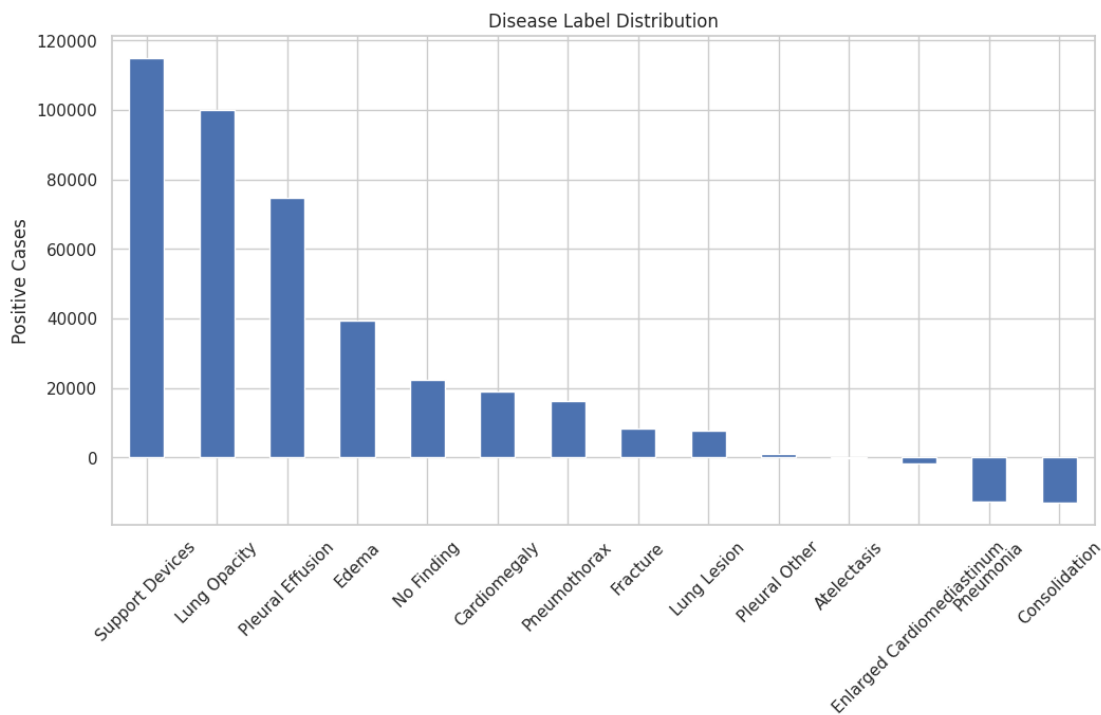


Figure 3.1

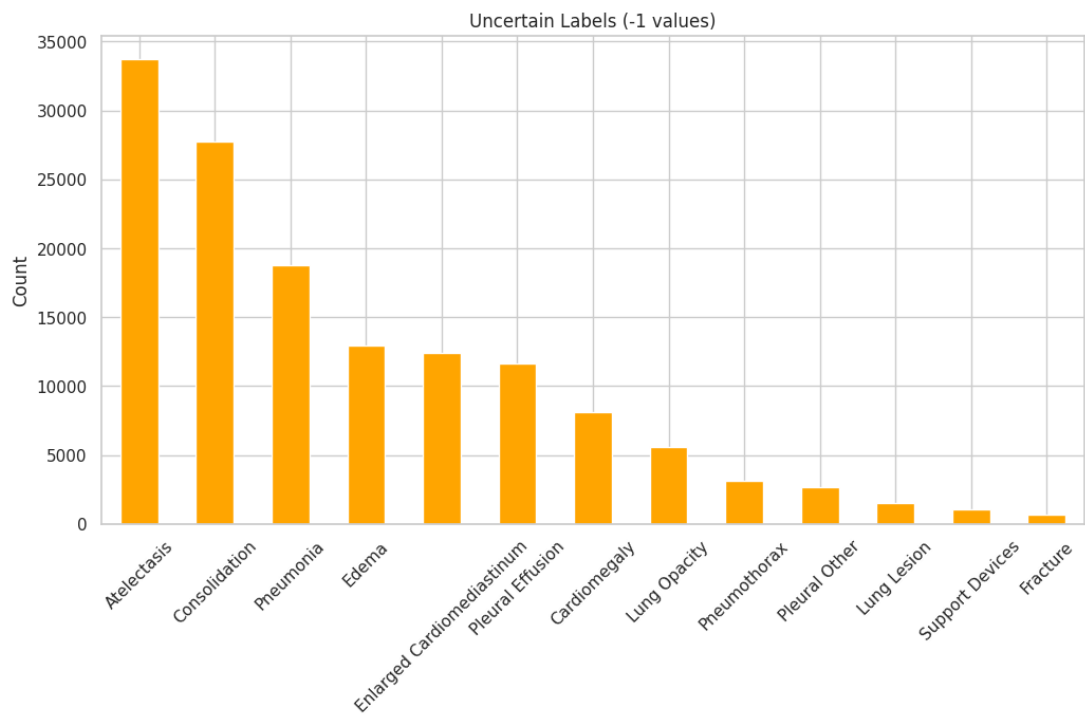


Figure 3.2

These patterns are consistent with known clinical dependencies and indicate that multiple pathologies often appear together in chest X-rays.

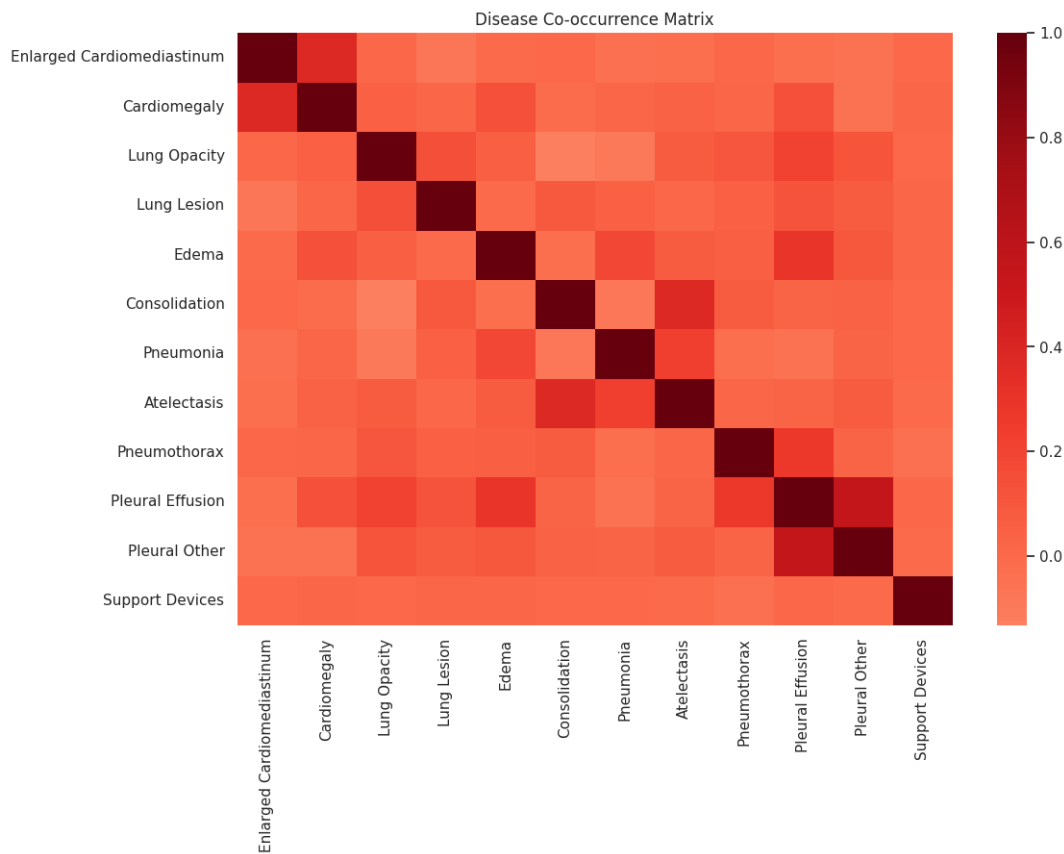


Figure 3.3

The CheXpert dataset contains a significant number of **missing and uncertain labels** due to automatic extraction from radiology reports. Uncertain labels are encoded as -1 and reflect ambiguity rather than absence. Missing values are replaced with 0, while uncertain labels (-1) are mapped to 0.5. This preserves partial information while maintaining a consistent numerical representation suitable for clustering.

Outliers are mainly present in the *Age* feature. The Interquartile Range (IQR) method is used to remove extreme values. This reduces noise and prevents extreme values from distorting the clustering structure.

Several **inconsistencies** were identified, including invalid *Sex* values, presence of healthy patients (*No Finding = 1*), and non-informative technical features. Samples with unknown *Sex* are removed, healthy cases (*No Finding = 1*) are excluded, and non-clinical features such as image paths and view positions are discarded. These steps ensure a consistent and clinically relevant dataset aligned with the goal of disease phenotype clustering.

From the exploratory analysis, several important insights emerge:

- The dataset is highly imbalanced, which may influence clustering results and bias the represen-

tation of dominant conditions.

- Missing and uncertain labels are prevalent and must be explicitly addressed during preprocessing.
- Certain diseases exhibit strong co-occurrence patterns, suggesting the presence of latent structures suitable for phenotype discovery.
- Weak overall correlations but meaningful local relationships indicate that non-linear methods (e.g., deep feature extraction) are appropriate.

Overall, the EDA highlights the complexity and multimodal nature of the dataset, justifying the need for robust feature extraction and fusion techniques before applying clustering algorithms.

3.1.3 Training

The training pipeline was designed to integrate high-dimensional visual information with structured clinical data to support multimodal phenotype discovery. The process followed a structured preprocessing workflow, followed by the application of multiple clustering algorithms representing centroid-based, density-based, hierarchical, graph-based, and probabilistic approaches.

3.1.3.1 Training pipeline

The preprocessing stage focused on transforming heterogeneous raw data into a model-ready matrix:

- **Tabular Cleaning:** Raw clinical variables were cleaned, and non-informative technical features (e.g., image paths and view positions) were discarded to ensure focus on clinical relevance [3.1.2](#).
- **Visual Feature Extraction:** Image features were extracted using a Vision Transformer (ViT)² by Google, providing a dense representation of the chest X-ray findings.
- **Data Fusion:** Tabular features were merged with the ViT-extracted image embeddings.
- **Dimensionality Reduction:** Dimensionality reduction was introduced in the second and third experiments. The fused multimodal features were standardized using StandardScaler and reduced using Principal Component Analysis (PCA) with 50 components to reduce noise and improve clustering stability.

²<https://huggingface.co/google/vit-base-patch16-224-in21k>

3.1.3.2 Clustering Algorithms and Parameter Selection

We investigated five algorithms to capture different structures in the latent space:

- **K-Means:** A centroid-based approach selected for its efficiency and clear partitioning of the feature space.
- **HDBSCAN:** A density-based algorithm chosen for its ability to identify clusters of varying shapes and handle noise (outliers).
- **Hierarchical Clustering (HC):** An agglomerative approach used to understand the nested relationships between patient subgroups.
- **Spectral Clustering:** A graph-based approach selected to capture possible non-linear structures in the latent space by grouping samples according to their similarity relationships.
- **Gaussian Mixture Model (GMM):** A probabilistic clustering approach used to model the data as a mixture of Gaussian distributions and to allow soft cluster assignments.

To determine the optimal number of clusters for K-Means, we utilized the Elbow Method. As shown in the plot 3.4, the most significant drop in inertia occurs from $k = 2$ to $k = 3$. Beyond $k = 3$, the curve continues to decrease but at a significantly more gradual rate, suggesting that three clusters provide the best balance between compression and descriptive power.

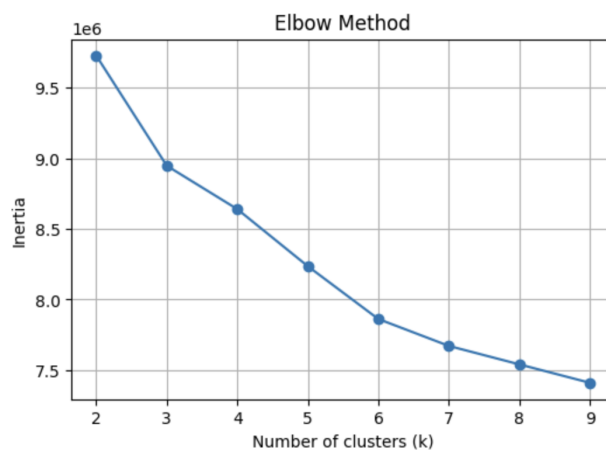


Figure 3.4: The elbow method for K-means

Based on the hierarchical clustering dendrogram 3.5, a cut around distance 250–300 produces three main branches. Therefore, the dendrogram suggests $k = 3$ as a reasonable number of clusters. This is consistent with the K-Means elbow result, which also indicated three clusters.

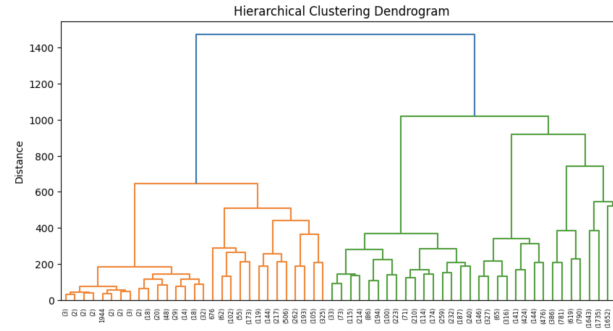


Figure 3.5: Hierarchical clustering dendrogram used for selecting the number of clusters

3.1.4 Results

We assessed the validity of the discovered clusters using three internal evaluation metrics that do not rely on ground-truth labels, which is essential for unsupervised phenotype discovery.

The following metrics were used to evaluate the clustering results:

- **Silhouette Coefficient:** Used to measure how similar an object is to its own cluster compared to other clusters.
- **Davies-Bouldin Index:** Used to evaluate the average similarity between each cluster and its most similar one; lower values indicate better separation.
- **Calinski-Harabasz Index:** Evaluates the ratio of between-cluster dispersion and within-cluster dispersion.

Table 3.1: Comparison of Clustering Algorithm Performance Metrics

Algorithm	n_clusters	noise_points	Silhouette	Davies-Bouldin	Calinski-Harabasz
HDBSCAN	2	1590	0.179	2.195	1073.59
HC	3	0	0.053	3.431	1247.296
KMeans	3	0	0.072	3.047	1434.13

In addition to these metrics, we also report the number of **noise points**, especially for density-based methods such as HDBSCAN. This is important because HDBSCAN is not forced to assign every sample to a cluster and can identify uncertain or outlier cases as noise.

Experiment 1: Baseline clustering without PCA

The first experiment was performed on the original multimodal feature space, without applying dimensionality reduction. Three clustering algorithms were compared: HDBSCAN, Hierarchical Clustering (HC), and K-Means.

Table 3.1 compares the performance of the baseline clustering approaches. HDBSCAN achieved the highest Silhouette Score of 0.179, suggesting that its clusters were better separated compared to HC and K-Means. It also obtained the lowest Davies-Bouldin Index of 2.195, further indicating better separation between the identified clusters.

However, HDBSCAN identified only 2 clusters and classified 1,590 samples as noise points. This means that, although the discovered clusters were more distinct, a considerable portion of the dataset was not assigned to any cluster. In contrast, HC and K-Means were forced to assign all samples to one of 3 clusters, resulting in 0 noise points.

K-Means achieved the highest Calinski-Harabasz score of 1434.13, followed by HC with 1247.30 and HDBSCAN with 1073.59. This is expected because K-Means directly optimizes compact centroid-based partitions, which can lead to stronger performance on this metric even when the Silhouette and Davies-Bouldin scores suggest weaker separation.

Overall, the first experiment showed that HDBSCAN produced the most clearly separated clusters according to Silhouette and Davies-Bouldin scores, but at the cost of leaving many samples as noise. K-Means provided a complete partitioning of the dataset and achieved the strongest Calinski-Harabasz score, but its lower Silhouette score suggests that the resulting clusters were less clearly separated.

Experiment 2: PCA, HDBSCAN tuning, and additional algorithms

In the second experiment, the clustering pipeline was improved in three ways. First, Principal Component Analysis (PCA) was applied to reduce the dimensionality of the multimodal feature space. This step was introduced to reduce noise, improve computational efficiency, and make the clustering algorithms more stable in a lower-dimensional representation.

Second, HDBSCAN was tuned by testing different hyperparameter configurations. The best reported configuration used a minimum cluster size of 90, minimum samples equal to 1, and the EOM cluster selection method. Third, two additional clustering algorithms were added: Spectral Clustering and Gaussian Mixture Models (GMM). This allowed us to compare centroid-based, density-based, hierarchical, graph-based, and probabilistic clustering approaches.

Table 3.2: Clustering validation metrics after PCA, HDBSCAN tuning, and additional algorithms.

Algorithm	Clusters	Noise	Silhouette	DBI	CHI
HDBSCAN	2	1456	0.209	1.888	1577.55
HC	3	0	0.102	2.520	1556.95
K-Means	3	0	0.098	2.605	1950.97
Spectral Clustering	3	0	0.096	2.335	1533.18
GMM	3	0	0.066	3.131	1591.96

The results show that HDBSCAN remained the best-performing method according to the Silhouette Score, achieving a value of 0.209. This is higher than the baseline result without PCA, where HDBSCAN achieved a Silhouette Score of 0.179. In addition, the number of noise points was reduced from 1,590 to 1,456. This indicates that PCA and hyperparameter tuning improved the HDBSCAN result by assigning more samples to meaningful clusters while also increasing cluster separation.

K-Means again achieved the highest Calinski-Harabasz Index, with a score of 1950.97. However, its Silhouette Score remained low at 0.098, suggesting that the clusters were not strongly separated despite being compact according to the CHI metric.

Spectral Clustering obtained a Silhouette Score similar to K-Means, but it achieved a better Davies-Bouldin Index of 2.335 compared to K-Means with 2.605. This suggests that Spectral Clustering may capture some non-linear structure in the data better than K-Means, although the improvement was not large.

GMM obtained the lowest Silhouette Score, 0.066, and the highest Davies-Bouldin Index, 3.131, indicating weaker cluster separation. This suggests that the Gaussian distribution assumption may not fit the structure of the multimodal feature space well.

Overall, the second experiment confirmed that HDBSCAN remained the strongest algorithm in terms of cluster separation, especially after hyperparameter tuning reduced the number of noise points. However, K-Means remained competitive due to its simplicity, full assignment of all samples, and strong Calinski-Harabasz score.

Experiment 3: External validation on NIH ChestX-ray14

To further evaluate the robustness of the clustering pipeline, a third experiment was conducted on a second dataset, NIH ChestX-ray14. The same experimental setup from Experiment 2 was applied, including PCA, the same clustering algorithms, and the same internal validation metrics.

Table 3.3: Clustering validation metrics on the NIH ChestX-ray14 dataset.

Algorithm	Clusters	Noise	Silhouette	DBI	CHI
HDBSCAN	2	5338	0.108	1.840	28.78
Spectral Clustering	3	0	0.102	2.586	681.33
K-Means	3	0	0.095	2.665	957.20
Hierarchical Clustering	3	0	0.089	3.118	728.19
GMM	3	0	0.052	3.421	625.61

The results on NIH ChestX-ray14 show a similar trend to the previous experiments, but with generally lower clustering quality. HDBSCAN again achieved the highest Silhouette Score, 0.108, and the lowest Davies-Bouldin Index, 1.840, indicating that it produced the most separated clusters among

the tested methods. However, it classified 5,338 samples as noise, which suggests that a large part of the NIH dataset did not fit into dense and well-defined groups.

K-Means achieved the highest Calinski-Harabasz Index, with a value of 957.20, followed by Hierarchical Clustering and Spectral Clustering. As in the previous experiments, this indicates that K-Means produced compact clusters, but its lower Silhouette Score suggests that the separation between clusters was still limited.

Spectral Clustering performed slightly better than K-Means in terms of Silhouette Score, 0.102 compared to 0.095, but K-Means achieved a stronger Calinski-Harabasz score. This suggests that Spectral Clustering may produce slightly better separated clusters, while K-Means produces more compact groups.

GMM again obtained the weakest results, with the lowest Silhouette Score and the highest Davies-Bouldin Index. This confirms that the probabilistic Gaussian assumption was not well suited for the extracted multimodal feature representation.

Overall, the NIH ChestX-ray14 experiment supports the findings from the CheXpert experiments. HDBSCAN consistently produced the best cluster separation, but it also identified many samples as noise. K-Means remained a strong practical baseline because it assigned all samples to clusters and achieved the highest Calinski-Harabasz scores. However, the generally low Silhouette Scores across all methods suggest that disease phenotype discovery in chest X-ray datasets is challenging, and the underlying patient groups may not be clearly separable using the current feature representation alone.

3.1.5 Summary of results

Across the three experiments, HDBSCAN was the most effective method according to the Silhouette Score and Davies-Bouldin Index, suggesting that it was better at identifying well-separated and dense patient subgroups. Its main limitation was the presence of many noise points, especially on the NIH ChestX-ray14 dataset.

K-Means consistently achieved the highest Calinski-Harabasz Index, which indicates compact clusters. However, its lower Silhouette Scores suggest that these clusters may overlap in the feature space. Therefore, K-Means is useful as a simple and stable baseline, but its clusters should be interpreted carefully.

Spectral Clustering showed competitive results after PCA, especially in terms of Davies-Bouldin Index and Silhouette Score, but it did not clearly outperform HDBSCAN or K-Means. Hierarchical Clustering performed similarly to K-Means but with slightly weaker results. GMM was the least effective method across both datasets, indicating that the multimodal feature space does not follow a

simple Gaussian mixture structure.

These experiments show that PCA and hyperparameter tuning improved the clustering pipeline, especially for HDBSCAN, by increasing its Silhouette Score and reducing the number of noise points compared to the baseline experiment. The external validation on NIH ChestX-ray14 further demonstrated that similar general trends are preserved across datasets, although clustering performance decreased on the second dataset. This suggests that the discovered disease phenotypes are partially dataset-dependent and that further improvements in feature extraction or multimodal representation learning may be necessary.

3.1.6 Validation

The primary objective of the experiments was to evaluate whether the extracted multimodal feature space allows the discovery of meaningful patient phenotypes. To complement the quantitative internal validation metrics, we also applied t-Distributed Stochastic Neighbor Embedding (t-SNE) to visually inspect the clustering behavior in a two-dimensional projection.

Three validation experiments were analyzed visually:

- **Experiment 1:** Baseline clustering on the original feature space, without PCA.
- **Experiment 2:** Clustering after PCA, HDBSCAN hyperparameter tuning, and the addition of Spectral Clustering and Gaussian Mixture Models.
- **Experiment 3:** External validation on the NIH ChestX-ray14 dataset using the same setup as Experiment 2.

Experiment 1: Baseline t-SNE visualization

Figure 3.6 shows the t-SNE visualization for the first experiment. As observed in the projections, the latent feature space is characterized by one clearly distinct sub-population, represented by the isolated island on the far left, and a much larger, dense, continuous central mass. The three baseline algorithms handled this distribution differently:

- **K-Means ($k = 3$):** K-Means produced a clear and complete partitioning of the feature space. It isolated the distinct far-left subgroup and divided the large central mass into two contiguous regions. However, because the central region appears continuous rather than naturally separated, this split may reflect an imposed algorithmic boundary rather than a true clinical separation.

- **Hierarchical Clustering** ($k = 3$): Similar to K-Means, Hierarchical Clustering successfully separated the isolated far-left group. However, it struggled with the dense central mass. The assignments for the remaining two clusters appear highly intermixed and overlapping, visually confirming the lower Silhouette score compared to K-Means.
- **HDBSCAN**: The density-based approach highlights the non-uniform structure of the feature space. It identified two main clusters: a smaller dense subgroup on the far left and a larger cluster covering most of the central and right-hand regions. At the same time, many scattered samples were labelled as noise, especially around the boundaries and lower-density parts of the distribution. This is consistent with the quantitative results, where HDBSCAN achieved the best separation metrics while still leaving 1,590 samples unassigned.

Overall, the first t-SNE visualization suggests that the original feature space contains one clearly separable subgroup, while the remaining samples form a continuous region that is difficult to divide into clearly distinct phenotypes.

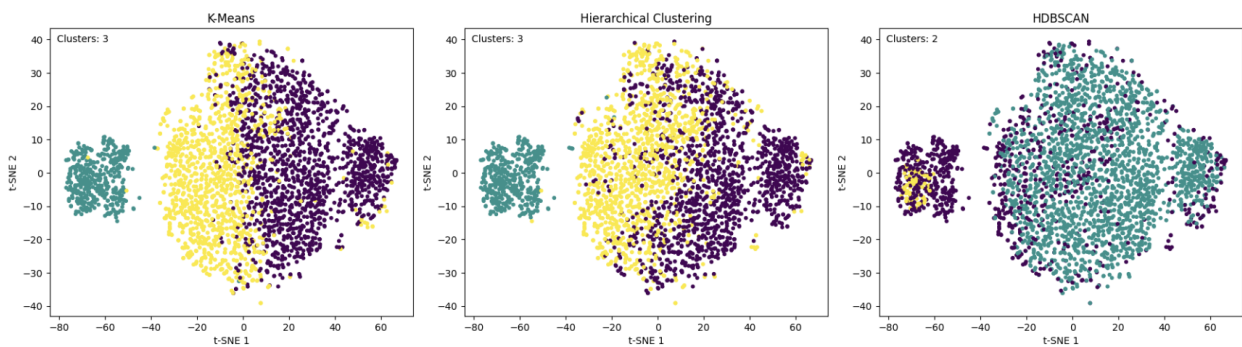


Figure 3.6: t-SNE visualization for Experiment 1: baseline clustering without PCA.

Experiment 2: t-SNE visualization after PCA and additional algorithms

Figure 3.7 presents the t-SNE visualization for the second experiment, where PCA was applied before clustering, HDBSCAN was tuned, and two additional algorithms were included: Spectral Clustering and Gaussian Mixture Models.

Compared to the first experiment, the PCA-based setup produced a more stable representation and reduced the number of HDBSCAN noise points. However, the visualization still shows that the data does not naturally separate into three fully distinct groups. Instead, most samples remain concentrated in a dense and continuous region.

- **K-Means** ($k = 3$): K-Means continued to provide a clean partitioning of the data. The clusters

appear spatially organized in the t-SNE plot, but the transitions between groups remain gradual. This suggests that K-Means creates useful and interpretable divisions, although these divisions may still be influenced by the algorithm's tendency to impose spherical clusters.

- **Hierarchical Clustering ($k = 3$):** Hierarchical Clustering produced similar behavior to the first experiment. It was able to identify broad group structures, but the central region remained visually overlapping. This indicates that the nested structure captured by HC is not strongly separated in the projected latent space.
- **HDBSCAN:** After hyperparameter tuning, HDBSCAN reduced the number of noise points compared to the first experiment while still identifying the densest regions of the data. In the visualization, it forms one dominant cluster over most of the central mass and a smaller cluster on the far left, while scattered low-density samples remain labelled as noise. This suggests that PCA and tuning improved the usability of HDBSCAN by allowing more samples to be assigned to clusters while preserving its ability to detect uncertain or low-density cases.
- **Spectral Clustering ($k = 3$):** Spectral Clustering was included to capture possible non-linear structures in the latent space. In the visualization, it assigns most of the central mass to one dominant cluster while separating smaller peripheral regions into the remaining groups. This suggests that the method is sensitive to graph-based similarity structure, although the resulting partition is still not strongly separated in the overall feature space.
- **Gaussian Mixture Model ($k = 3$):** GMM provided a probabilistic clustering perspective by modelling the feature space as a mixture of Gaussian distributions. However, the t-SNE projection suggests that the data does not clearly follow well-separated Gaussian components. This supports the weaker quantitative results obtained by GMM, especially its lower Silhouette score and higher Davies-Bouldin Index.

Overall, the second experiment shows that PCA and HDBSCAN tuning improved the clustering setup, especially by reducing the number of noise points. Nevertheless, the visual structure remains only partially separable, suggesting that the discovered phenotypes should be interpreted as latent patient subgroups rather than sharply separated clinical classes.

Experiment 3: t-SNE visualization on NIH ChestX-ray14

Figure 3.8 shows the t-SNE visualization for the third experiment, performed on the NIH ChestX-ray14 dataset using the same setup as in Experiment 2. This experiment was used as an external validation

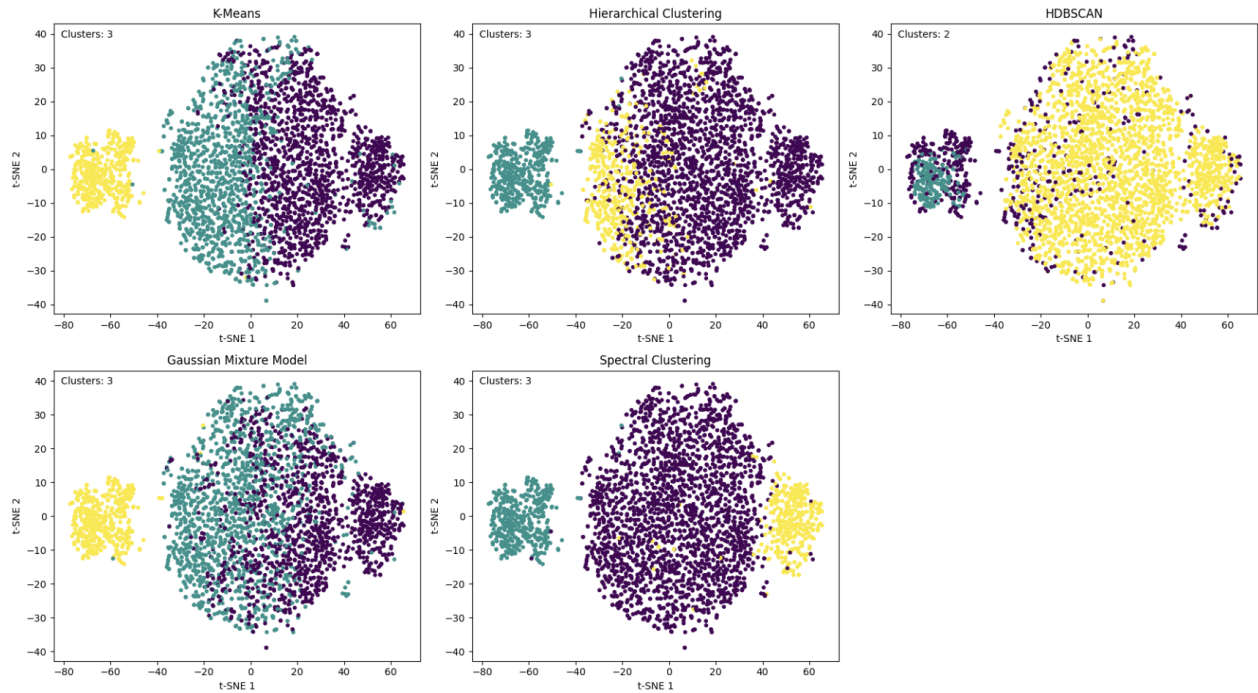


Figure 3.7: t-SNE visualization for Experiment 2: clustering after PCA, HDBSCAN tuning, Spectral Clustering, and GMM.

step to evaluate whether the same clustering behavior appears on a different chest X-ray dataset.

The NIH ChestX-ray14 results show a generally more challenging clustering structure. Compared to CheXpert, the projected samples appear less clearly separated, which is consistent with the lower Silhouette scores obtained across all algorithms. This indicates that the NIH feature space contains weaker natural cluster boundaries under the current representation.

- **K-Means ($k = 3$):** K-Means again produced a complete partition of the dataset into three clusters. However, the t-SNE visualization suggests that these clusters overlap more strongly than in the CheXpert experiments. This supports the interpretation that K-Means provides a stable baseline, but not necessarily clinically distinct phenotypes.
- **Hierarchical Clustering ($k = 3$):** Hierarchical Clustering also assigned all samples to clusters, but the visual separation between groups remained weak. The overlap between clusters indicates that the hierarchical structure in the NIH dataset is less clearly defined.
- **HDBSCAN:** HDBSCAN obtained the best Silhouette Score and Davies-Bouldin Index on NIH ChestX-ray14, but the visualization shows that it labelled a very large proportion of the samples as noise, retaining only limited dense structure as clusters. This suggests that only a relatively small subset of the NIH data forms compact and reliable groups, while most samples lie in

ambiguous or weakly separated regions of the feature space.

- **Spectral Clustering ($k = 3$):** Spectral Clustering showed competitive behavior and achieved a slightly higher Silhouette Score than K-Means on the NIH dataset. This suggests that graph-based similarity relationships may capture some structure in the data, although the improvement remains limited.
- **Gaussian Mixture Model ($k = 3$):** GMM again produced the weakest visual and quantitative clustering performance. The overlap between clusters suggests that the NIH feature representation does not fit well into a mixture of simple Gaussian components.

Overall, the third experiment confirms that the general trends observed on CheXpert are also present on NIH ChestX-ray14. HDBSCAN is the most selective and identifies the most separated dense regions, but at the cost of many noise points. K-Means and Spectral Clustering provide complete partitions of the data, but their clusters are less clearly separated. The lower overall scores on NIH ChestX-ray14 suggest that phenotype discovery is dataset-dependent and that the current multimodal representation may require further refinement for stronger generalization.

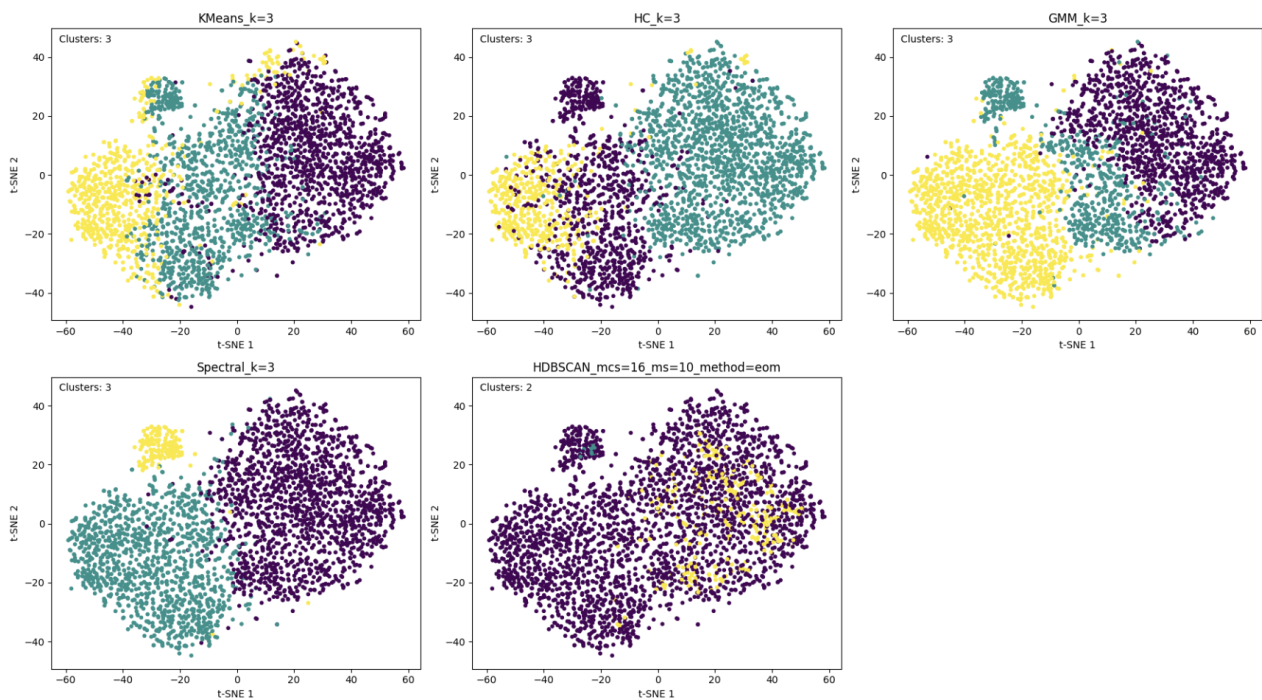


Figure 3.8: t-SNE visualization for Experiment 3: external validation on the NIH ChestX-ray14 dataset.

Cluster interpretation for the best HDBSCAN model

Since HDBSCAN achieved the strongest performance in Experiment 2 according to the Silhouette Score and Davies-Bouldin Index, we further analyzed the clinical composition of its discovered clusters. For each cluster, we computed the prevalence of each disease label and compared it with the global prevalence in the full non-noise dataset. This comparison was expressed using the lift score, where values above 1 indicate that a disease is overrepresented in a specific cluster compared to the overall dataset.

Table 3.4: Most characteristic disease labels for the HDBSCAN clusters in Experiment 2.

Cluster	Disease	Cluster Size	Disease Count	Cluster Prev.	Lift
0	Edema	14998	5157	0.344	1.073
0	Support Devices	14998	8581	0.572	1.049
0	Pneumothorax	14998	1810	0.121	1.032
0	Lung Opacity	14998	8508	0.567	1.023
0	Atelectasis	14998	5166	0.344	1.020
1	Pleural Other	1576	106	0.067	2.186
1	Lung Lesion	1576	148	0.094	1.810
1	Enlarged Cardiomedastinum	1576	224	0.142	1.247
1	Pneumonia	1576	220	0.140	1.188
1	Cardiomegaly	1576	273	0.173	1.015

The interpretation shows that **Cluster 0** represents the dominant patient group, containing 14,998 samples. Its most frequent labels are common findings such as *Support Devices*, *Lung Opacity*, *Edema*, and *Atelectasis*. However, the lift values are close to 1, meaning that these labels are only slightly more frequent in this cluster than in the overall dataset. Therefore, Cluster 0 can be interpreted as a broad, heterogeneous phenotype that captures the main distribution of common chest X-ray findings rather than a highly specific disease subgroup.

In contrast, **Cluster 1** is much smaller, containing 1,576 samples, but it shows stronger disease enrichment. The highest lift values are observed for *Pleural Other* with a lift of 2.186 and *Lung Lesion* with a lift of 1.810. This means that these findings are considerably more prevalent in Cluster 1 than in the global dataset. The cluster also shows moderate enrichment for *Enlarged Cardiomedastinum* and *Pneumonia*. Therefore, Cluster 1 can be interpreted as a more specific phenotype associated with less frequent but more distinctive thoracic abnormalities.

Overall, this analysis suggests that the best HDBSCAN model separates the dataset into one large heterogeneous cluster containing common radiological findings and one smaller, more clinically specific cluster enriched in pleural and lesion-related abnormalities. This supports the usefulness of HDBSCAN for phenotype discovery, since it does not only optimize internal clustering metrics but

also reveals interpretable differences in disease-label composition between patient subgroups.

Cluster interpretation for Spectral Clustering on NIH ChestX-ray14

Since Spectral Clustering provided a complete partition of the NIH ChestX-ray14 dataset into three clusters, we further analyzed the disease-label composition of each cluster. As in the CheXpert experiment, disease prevalence inside each cluster was compared with the global disease prevalence using the lift score. A lift value greater than 1 indicates that a disease is overrepresented in that cluster compared to the full dataset.

Table 3.5: Most characteristic disease labels for Spectral Clustering on NIH ChestX-ray14.

Cluster	Disease	Cluster Size	Disease Count	Cluster Prev.	Lift
0	Fibrosis	4580	244	0.053	1.483
0	Hernia	4580	28	0.006	1.423
0	Pleural Thickening	4580	372	0.081	1.255
0	Nodule	4580	672	0.147	1.164
0	Pneumothorax	4580	384	0.084	1.148
1	Edema	3367	262	0.078	2.063
1	Consolidation	3367	430	0.128	1.544
1	Atelectasis	3367	944	0.280	1.202
1	Effusion	3367	934	0.277	1.137
1	Pneumonia	3367	93	0.028	1.123
2	Hernia	430	4	0.009	2.165
2	Pneumothorax	430	47	0.109	1.496
2	Emphysema	430	24	0.056	1.347
2	Cardiomegaly	430	29	0.067	1.329
2	Pleural Thickening	430	35	0.081	1.258

The interpretation shows that **Cluster 0** is the largest group, containing 4,580 samples. It is mainly characterized by elevated prevalence of *Fibrosis*, *Hernia*, *Pleural Thickening*, *Nodule*, and *Pneumothorax*. However, most lift values are moderate, suggesting that this cluster represents a broad phenotype containing chronic or structural thoracic abnormalities rather than one highly specific disease pattern.

Cluster 1, with 3,367 samples, shows stronger enrichment for acute pulmonary findings. The most characteristic label is *Edema*, with a lift of 2.063, followed by *Consolidation*, *Atelectasis*, *Effusion*, and *Pneumonia*. This suggests that Cluster 1 may correspond to a phenotype associated with fluid accumulation, inflammatory findings, and lung collapse-related abnormalities.

Cluster 2 is the smallest group, containing only 430 samples, but it shows the strongest enrichment for some rare findings. In particular, *Hernia* has a lift of 2.165, while *Pneumothorax*, *Emphysema*, *Cardiomegaly*, and *Pleural Thickening* are also overrepresented. Because of its small size and higher lift values, this cluster may represent a more specific subgroup of patients with rarer or more distinctive

radiological patterns.

Overall, the Spectral Clustering interpretation on NIH ChestX-ray14 suggests three clinically different patient subgroups: a large cluster associated with chronic or structural abnormalities, a second cluster enriched in acute pulmonary and fluid-related findings, and a smaller cluster enriched in rarer thoracic conditions. This supports the usefulness of cluster interpretation based on label prevalence and lift, since it provides clinical meaning beyond the internal validation metrics alone.

Chapter 4

Discussion

The use of medical imaging data and machine learning methods raises important ethical, clinical, and societal questions. Although the proposed approach is exploratory and is not intended for direct clinical decision-making, the discovered clusters may influence how patient subgroups and disease patterns are interpreted. Therefore, it is important to examine not only the technical performance of the pipeline, but also its limitations, possible biases, reliability, interpretability, and potential impact in real-world healthcare contexts.

In this project, multimodal clustering is used to identify hidden patterns from chest X-ray images and structured patient-related information. Such an approach can support medical research by revealing possible patient phenotypes, but it also carries risks if the results are overinterpreted or used without clinical validation. For this reason, the following section discusses the strengths and limitations of the proposed method, the ethical issues related to data and modeling, and the possible business and societal impact of applying such systems in healthcare.

4.1 Strengths and Limitations

The main strength of the proposed approach is that it combines multiple sources of information instead of relying only on images or only on tabular labels. This makes the representation more informative and closer to real clinical reasoning, where radiologists consider both imaging findings and patient-related information. Another strength is the comparison of multiple clustering algorithms, which gives a broader view of how different methods behave on the same multimodal feature space. The external validation on NIH ChestX-ray14 also strengthens the analysis by testing the pipeline on an additional dataset.

However, the approach also has several limitations. First, the image features were extracted using

a pretrained model that was not specifically trained for phenotype discovery. As a result, the learned representation may not fully capture the subtle visual differences between disease subtypes. Second, the fusion strategy was based on simple concatenation of image and tabular features, which may not model complex interactions between modalities. Third, the evaluation relied mainly on internal clustering metrics and label-based interpretation, since no ground-truth phenotype labels were available. Finally, the low Silhouette Scores observed across experiments suggest that the discovered clusters are not sharply separated and should be interpreted carefully.

A SWOT analysis of the proposed approach is presented below:

- **Strengths:** The pipeline uses both chest X-ray images and structured tabular data, compares multiple clustering algorithms, and provides interpretable cluster analysis using disease prevalence and lift scores.
- **Weaknesses:** The feature fusion method is simple, the pretrained image encoder may not be optimal for this task, and the clustering quality remains limited due to overlapping and imbalanced medical data.
- **Opportunities:** Future work could improve the representation using deep multimodal learning, autoencoders, contrastive learning, or specialized medical image encoders. The approach could also be extended to include additional clinical variables, such as laboratory results or clinical reports.
- **Threats:** The discovered clusters may be dataset-dependent and may not generalize to other hospitals or patient populations. In addition, biases in the original datasets, missing labels, and uncertain annotations may influence the resulting phenotypes.

4.2 Ethical Considerations

One important ethical aspect is related to the datasets used in this study. Chest X-ray datasets such as CheXpert and NIH ChestX-ray14 are large and useful for research, but they may still contain biases. For example, the distribution of diseases, age groups, sex, acquisition devices, and hospital settings may not fully represent all patient populations. If some groups are underrepresented, the discovered clusters may describe the dominant groups better than rare or minority groups. This can limit the generalization of the results to other hospitals or populations.

Data quality is another important issue. The diagnostic labels in these datasets are often extracted automatically from radiology reports, which means that they may contain noise, missing values, or

uncertain annotations. In this project, missing and uncertain labels were handled during preprocessing, but this does not completely remove the risk of incorrect or ambiguous information. Since clustering is sensitive to the input representation, low-quality labels or noisy features may influence the discovered phenotypes.

Privacy must also be considered when working with medical images and patient-related information. Even if public datasets are usually anonymized, medical data remain sensitive. Any future use of such a system in a real clinical environment would require strict protection of patient identity, secure data storage, controlled access, and compliance with medical data protection regulations.

From the methods perspective, transparency and interpretability are essential. Clustering algorithms can group samples based on mathematical similarity, but these groups do not automatically correspond to clinically meaningful disease categories. To reduce this risk, the clusters were interpreted using disease prevalence and lift scores. This makes the results more understandable, but expert clinical validation would still be necessary before making any medical conclusions.

Fairness is also relevant because the clustering process may reflect existing biases in the data. For example, if certain diseases or demographic groups dominate the dataset, some algorithms may create clusters that mostly represent frequent findings, while rare conditions may be treated as noise or merged into larger groups. This was especially visible for HDBSCAN, which identified many noise points. In a clinical context, such samples should not be ignored, because they may correspond to rare or complex patient cases.

Another ethical risk is automation. The proposed approach should not be used as an automatic diagnostic tool. The clusters are exploratory and should support analysis, not replace medical decision-making. Incorrect interpretation of clusters could lead to misleading conclusions about patient similarity or disease subtypes. Therefore, any deployment of this type of system should include human supervision, especially from medical experts.

Finally, the reliability of the results must be considered. The internal validation metrics showed relatively low Silhouette Scores, meaning that the clusters are not strongly separated. This indicates that the proposed solution is useful for exploratory analysis, but not reliable enough for direct clinical decisions. The results should be interpreted as possible disease patterns rather than confirmed medical phenotypes.

4.3 Business and Societal Impact

From a business and industry perspective, a multimodal clustering system could be useful in healthcare analytics, hospital research, clinical decision support, and medical AI development. For example, hospitals or research organizations could use such a pipeline to explore large collections of chest X-rays and identify patient subgroups with similar imaging and clinical characteristics. This could support retrospective studies, cohort selection, and hypothesis generation for future medical research.

A potential benefit for healthcare organizations is improved understanding of disease heterogeneity. Instead of analyzing all patients with the same diagnosis as one homogeneous group, clustering can reveal subgroups with different combinations of findings. This may help researchers study disease progression, compare patient profiles, or identify groups that require closer clinical attention.

The approach could also have operational benefits. In large hospitals, radiology datasets are difficult to analyze manually at scale. Automated clustering can help organize large image collections, summarize common patterns, and highlight unusual or rare cases. This could reduce the time needed for exploratory analysis and support better use of existing medical data.

Economically, such systems may contribute to more efficient research and resource allocation. If patient subgroups are better understood, healthcare organizations may be able to design more targeted studies, improve prioritization, or support personalized treatment planning in the long term. However, these benefits depend on strong validation, clinical acceptance, and integration with existing hospital workflows.

The societal impact of this work is connected to personalized medicine. By discovering hidden patient subgroups, multimodal clustering may contribute to a better understanding of why patients with the same diagnosis can present different symptoms, imaging patterns, or outcomes. In the long term, this could support more individualized healthcare strategies.

However, there are also risks. If the system is used without proper validation, it could reinforce dataset biases or produce misleading groupings. Incorrectly interpreted clusters could affect research conclusions or clinical decisions. Therefore, the proposed solution should be seen as an exploratory tool, not as a final decision-making system.

Chapter 5

Conclusion and future work

This project investigated the use of multimodal clustering for disease-phenotype discovery from chest X-ray data. The main goal was to group similar patients or images by combining visual information extracted from radiographs with structured tabular information such as age, sex, and diagnostic observations. Instead of solving a supervised classification problem, the study focused on discovering hidden patient subgroups using unsupervised learning.

The proposed pipeline included data preprocessing, image feature extraction using a pretrained Vision Transformer, fusion of image and tabular features, optional dimensionality reduction using PCA, and the application of several clustering algorithms. The investigated algorithms included K-Means, HDBSCAN, Hierarchical Clustering, Spectral Clustering, and Gaussian Mixture Models. The clusters were evaluated using internal validation metrics such as the Silhouette Coefficient, Davies-Bouldin Index, and Calinski-Harabasz Index. In addition, the clinical meaning of the clusters was analyzed using disease prevalence and lift scores.

The experimental results showed that HDBSCAN achieved the best performance in terms of Silhouette Score and Davies-Bouldin Index, suggesting that it was the most effective method for identifying dense and better-separated patient subgroups. However, a major limitation of HDBSCAN was that it labelled many samples as noise, especially on the NIH ChestX-ray14 dataset. K-Means was a strong and stable baseline, achieving the highest Calinski-Harabasz scores, but its clusters were less clearly separated according to the Silhouette metric. Spectral Clustering showed competitive results, especially after PCA, while Gaussian Mixture Models produced the weakest results, indicating that the multimodal feature space was not well described by simple Gaussian components.

An important conclusion of this work is that disease-phenotype discovery in chest X-ray datasets is challenging because medical data are highly heterogeneous, imbalanced, and often noisy. Patients with the same diagnosis may present different visual patterns, while different diseases may also co-

occur in the same image. Despite these challenges, the results show that multimodal clustering can reveal meaningful structures in the data. For example, the cluster interpretation step showed that some clusters were enriched in specific findings such as pleural abnormalities, lung lesions, edema, consolidation, or pneumothorax. This suggests that clustering can provide useful exploratory insights beyond standard disease labels.

Future work should focus on improving both the multimodal representation and the validation strategy. A first direction would be to replace simple feature concatenation with more advanced fusion methods, such as multimodal autoencoders, attention-based fusion, or contrastive multimodal learning. These methods could learn a more meaningful shared latent space before clustering. A second direction would be to fine-tune the image encoder on chest X-ray data, instead of using fixed pretrained features. This could help capture more disease-specific visual patterns.

Another possible improvement is to include stronger clinical validation. For example, the discovered clusters could be compared with expert radiologist annotations, patient outcomes, or additional clinical variables. Cluster stability analysis could also be used to test whether the same phenotypes appear consistently under different samples, random seeds, or preprocessing choices. Finally, future experiments could explore deep clustering methods that jointly learn the representation and cluster assignments, rather than applying clustering only after feature extraction.

Overall, this work demonstrates that multimodal clustering is a promising approach for exploring disease heterogeneity in chest X-ray datasets. Although the current results are exploratory and not directly suitable for clinical decision-making, they show that combining image and tabular data can reveal patient subgroups with different disease characteristics. This can support future research in medical data analysis, representation learning, and personalized healthcare.

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